

Tarun Kari

Berlin, Germany | Graduating Oct. 2026 | +49 15560711688 | tarunkari0411@gmail.com | [LinkedIn](#) | [GitHub](#)

EDUCATION

International University of Applied Sciences

Berlin, Germany

Master of Science in Data Science (GPA: 2.0/4.0)

Oct. 2024 – Oct. 2026

- **Relevant Coursework:** Programming with Python, Machine Learning, Data Science Use Case.

EXPERIENCE

Software Engineer

July 2022 – Aug. 2024

LTIMindtree

Bangalore, India

- Architected and developed **scalable backend services in Python using FastAPI/Pydantic-oriented patterns for agent workflows, forecasting, and replenishment-style use cases**, covering service boundaries, validation, API contracts, operational trade-offs, deployment, and reliability considerations.
- Designed **clean API and service contracts between platforms, agents, ML models, frontend applications, and customer systems such as ERP, BI, and planning tools**, emphasizing robustness, observability, reliability, reusable interfaces, retries, validation, and clear ownership boundaries.
- Worked with **PostgreSQL modeling, dbt-style transformations, data pipelines, caching, and retrieval patterns** to create consistent data flows that could scale across customer and analytical use cases; improved processed-data quality from approximately **85% to 97%**.
- Built backend services with **Python, FastAPI, Pydantic-style validation, asynchronous programming concepts, PostgreSQL, Redis, Docker, and mature CI/CD practices**, applying modular architecture, automated testing, logging, Git, Jenkins/GitHub workflows, and deployment automation.
- Delivered in **Agile, fast-moving teams through sprints, iterative shipping, and continuous feedback**, balancing speed with code quality, testability, maintainability, and production reliability while reducing repetitive manual verification by approximately **40%**.
- Worked with **AWS cloud infrastructure, Kubernetes concepts, and Infrastructure-as-Code with Terraform-style workflows**, contributing to deployment, monitoring, production troubleshooting, architecture discussions, and pragmatic trade-offs between scalability, complexity, and maintainability.

Software Engineer Intern

Mar. 2022 – June 2022

LTIMindtree

Bangalore, India

- Developed backend and data-processing components using **Python, Java, REST APIs, SQL, relational databases, validation, and reusable service patterns**, supporting integration and application workflows.
- Used **Git, automated testing, CI/CD, code reviews, debugging, documentation, and Agile/Scrum workflows** while collaborating with senior engineers on maintainable production-oriented software.

PROJECTS

NovaRAG | *Python, OpenAI, Anthropic, LangChain, FAISS, ChromaDB, RAGAS*

2026 – Present

- Built a **production-style RAG and agent platform** that ingests PDF, DOCX, HTML, TXT, JSON, and web content, retrieves relevant context, and generates grounded responses across configurable OpenAI, Anthropic, and local Hugging Face models.
- Implemented **tool/function-calling concepts, prompt engineering, hybrid retrieval, HyDE, query decomposition, metadata filtering, BM25/Cross-Encoder reranking, caching, provider abstractions, and conversation state** for flexible agent workflows.
- Added **RAGAS evaluation for faithfulness, answer relevancy, context precision/recall, PyTest, Pydantic, structured logging, Ruff, mypy, and configurable quality checks** to improve reliability, guardrail-oriented validation, and production readiness.

Sentinel Fraud Guard | *Python, FastAPI, PostgreSQL, Airflow, MLflow, Docker*

2026 – Present

- Built an **end-to-end fraud decision platform** that scores credit-card transactions through REST APIs and supports automated retraining, evaluation, model registration, and controlled promotion.
- Designed the architecture using **FastAPI for inference, PostgreSQL for persistence, Airflow for orchestration, MLflow for model tracking/registry, XGBoost with SMOTE, and Docker Compose for service deployment**.
- Added validation, health checks, logging, monitoring, model-quality gates, and persistent volumes to make the platform **observable, reproducible, fault-aware, and production-oriented**.

Zero-Downtime Predictive Maintenance | *Python, FastAPI, Kafka, PostgreSQL, PyTorch, Docker*

2026 – Present

- Built a **multimodal forecasting and predictive-maintenance platform** that analyzes industrial sensor signals and maintenance text to classify failure modes and estimate Remaining Useful Life.
- Combined **1D CNN/LSTM models with DistilBERT embeddings**, while Kafka handled event-driven ingestion, PostgreSQL/TimescaleDB stored time-series data, and FastAPI exposed inference services.
- Added structured logging, schema validation, service isolation, Dockerized deployment, and runtime diagnostics to improve **reliability, observability, failure analysis, and production deployment readiness**.

TECHNICAL SKILLS

Backend & APIs: Python, FastAPI, Pydantic, REST APIs, Async Programming Concepts, Microservices, API Contracts, Service Architecture

Agents & LLMs: OpenAI, Anthropic, LangChain, RAG, Tool Use, Function Calling, Prompt Engineering, Agent Workflows, Evaluation Frameworks, Guardrail Concepts

AI-First Engineering: Claude Code Concepts, AI-Assisted Coding, Plan/Execute/Review Workflows, Subagent Concepts, Prompt/Tool Iteration, Documentation Automation

Data Platform: PostgreSQL, Redis, dbt Familiarity, Data Pipelines, Caching, Retrieval, Semantic Layer Concepts, Airflow, TimescaleDB

ML & Forecasting: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, Forecasting, Classification, Deep Learning, NLP

Cloud & DevOps: AWS Fundamentals, Docker, Kubernetes Concepts, Terraform Familiarity, Git, GitHub Actions, Jenkins, CI/CD, IaC Concepts

Reliability: Logging, Monitoring, Observability, Health Checks, Root-Cause Analysis, Production Debugging, Error Handling, Performance Optimization

Product & Engineering: Product Mindset, Iterative Shipping, Technical Trade-offs, Code Reviews, Documentation, Agile/Scrum, Stakeholder Collaboration

Languages: English – Fluent, German – A2